

Docker 应用部署-RocketMQ

1 安装前准备

1.1 准备安装目录

```
1 mkdir /home/rocketmq
2 cd /home/rocketmq
```

1.2 下载rocketmq

下载 [rocketmq-all-4.9.3-bin-release.zip](http://archive.apache.org/dist/rocketmq/4.9.3/rocketmq-all-4.9.3-bin-release.zip) 文件，用于复制里面的 conf 目录文件。

可以手动下载后上传到服务器，然后使用unzip指令解压；

也可以wget命令下载后再解压，参考命令如下所示。

```
1 # 下载文件，如果服务器上下载缓慢建议手动下载后上传到服务器
2 wget http://archive.apache.org/dist/rocketmq/4.9.3/rocketmq-all-4.9.3-bin-release.zip
3 # 解压
4 unzip rocketmq-all-4.9.3-bin-release.zip
```

1.3 准备Broker服务配置

```
1 # 创建broker目录
2 mkdir broker
3 # 复制配置到broker
4 cp -r rocketmq-4.9.3/conf/ broker/conf/
```

```
[root@localhost rocketmq]# mkdir broker
[root@localhost rocketmq]# cp -r rocketmq-4.9.3/conf/ broker/conf/
[root@localhost rocketmq]# ls broker/conf/
2m-2s-async 2m-2s-sync 2m-noslave acl broker.conf dledger lo
[root@localhost rocketmq]#
```

修改broker/conf中的broker.conf文件，修改参考如下：

```
1 # 所属集群名称，如果节点较多可以配置多个
2 brokerClusterName = DefaultCluster
3 # broker名称，master和slave使用相同的名称，表明他们的主从关系
4 brokerName = broker-a
5 # 0表示Master，大于0表示不同的slave
6 brokerId = 0
7 # 表示几点做消息删除动作，默认是凌晨4点
8 deleteWhen = 04
9 # 在磁盘上保留消息的时长，单位是小时
10 fileReservedTime = 48
```

```

11 # 有三个值：SYNC_MASTER，ASYNC_MASTER，SLAVE；同步和异步表示Master和Slave之间同步数
    据的机制
12 brokerRole = ASYNC_MASTER
13 # 刷盘策略，取值为：ASYNC_FLUSH，SYNC_FLUSH表示同步刷盘和异步刷盘
14 # SYNC_FLUSH消息写入磁盘后才返回成功状态，ASYNC_FLUSH不需要
15 flushDiskType = ASYNC_FLUSH
16
17 # 设置broker节点所在服务器的ip地址（非常重要）
18 # 主从模式下，从节点会根据主节点的brokerIP2来同步数据，如果不配置，主从无法同步，
19 # brokerIP1设置为自己外网能访问的ip，服务器双网卡情况下必须配置，
20 # 比如阿里云这种，主节点需要配置ip1和ip2，从节点只需要配置ip1即可
21 brokerIP1 = 192.168.220.128
22 # nameServer地址，分号分割
23 namesrvAddr = 192.168.220.128:9876
24 # Broker对外服务的监听端口
25 listenPort = 10911
26 # 是否允许Broker自动创建Topic，建议线下开启，线上关闭
27 autoCreateTopicEnable = true
28 # 是否允许Broker自动创建订阅组，建议线下开启，线上关闭
29 autoCreateSubscriptionGroup = true
30 # linux开启epoll
31 useEpollNativeSelector = true
32 # 更多配置说明可参考
33 # https://blog.csdn.net/guyue35/article/details/105439325

```

```

[root@localhost rocketmq]# vi broker/conf/broker.conf
[root@localhost rocketmq]# cat broker/conf/broker.conf
# Licensed to the Apache Software Foundation (ASF) under one or more
# contributor license agreements. See the NOTICE file distributed with
# this work for additional information regarding copyright ownership.
# The ASF licenses this file to You under the Apache License, Version 2.0
# (the "License"); you may not use this file except in compliance with
# the License. You may obtain a copy of the License at
#
#     http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

brokerClusterName = DefaultCluster
brokerName = broker-a
brokerId = 0
deleteWhen = 04
fileReservedTime = 48
brokerRole = ASYNC_MASTER
flushDiskType = ASYNC_FLUSH
brokerIP1 = 192.168.220.128
namesrvAddr = 192.168.220.128:9876
listenPort = 10911
autoCreateTopicEnable = true
autoCreateSubscriptionGroup = true
useEpollNativeSelector = true
[root@localhost rocketmq]#

```

① 换成自己的IP

1.4 修改目录权限

修改权限前，先创建服务器需要的目录

```
1 mkdir -p broker/store broker/logs
2 mkdir -p namesrv/store namesrv/logs
```

容器中的用户ID和组ID分别是 3000 和 3000，如果不给宿主机目录授权那么容器将会运行不成功的

```
1 chown -R 3000:3000 ./
```

```
[root@localhost rocketmq]# chown -R 3000:3000 ./
[root@localhost rocketmq]# ll
总用量 26264
drwxr-xr-x. 3 3000 3000      18 7月  30 21:54 broker
drwxr-xr-x. 6 3000 3000    103 2月  22 01:25 rocketmq-4.9.3
-rw-r--r--. 1 3000 3000 26892203 7月  30 21:28 rocketmq-all-4.9.3-bin-release.zip
[root@localhost rocketmq]#
```

1.5 了解将要安装服务

Name Server:

名称服务充当路由消息的提供者。生产者或消费者能够通过名字服务查找各主题相应的Broker IP列表。

Broker Server:

代理服务器：消息中转角色，负责存储消息、转发消息。

代理服务器在RocketMQ系统中负责接收从生产者发送来的消息并存储、同时为消费者的拉取请求作准备。

代理服务器也存储消息相关的元数据，包括消费者组、消费进度偏移和主题和队列消息等。

Broker 启动时，（默认）实际上会监听3个端口：10909、10911、10912，这三个端口由Broker内部不同的组件使用，作用分别如下：

<http://www.tianshouzhi.com/api/tutorials/rocketmq/417>

- **remotingServer**: 监听 listenPort 配置项指定的监听端口，默认为 10911
- **fastRemotingServer**: 监听端口值 listenPort - 2，即默认为 10909
- **HAService**: 监听端口为值为 listenPort + 1，即默认为 10912，该端口用于 Broker 的主从同步

Console Server:

方便以图形化的方式运维MQ服务器情况，同时可以通过控制台对MQ服务器执行一些操作，如：创建主题、发布消息。

2 配置防火墙

注意：如果是云服务器端口放行需要到控制台页面去开放，一般情况操作的系统的防火墙是关闭的，不需要通过下面的命令行开放防火墙。

```
1 #开放端口
2 firewall-cmd --add-port 9876/tcp --add-port 10911/tcp --add-port 8280/tcp --
  permanent
3 #重新加载防火墙
4 firewall-cmd --reload
```

3 服务编排

在rocketmq目录下创建docker-compose.yml文件，并写入以下内容。

```
1 version: '3'
2 services:
3   rocketmq-namesrv:
4     image: apache/rocketmq:4.9.3
5     container_name: rocketmq-namesrv
6     ports:
7       - "9876:9876"
8     environment:
9       - JAVA_OPT_EXT=-server -Xms128m -Xmx128m -Xmn64m
10    volumes:
11      - /home/rocketmq/namesrv/logs:/home/rocketmq/logs
12      - /home/rocketmq/namesrv/store:/home/rocketmq/store
13    command: sh mqnamesrv
14    rocketmq-broker:
15      image: apache/rocketmq:4.9.3
16      container_name: rocketmq-broker
17      ports:
18        - "10909:10909"
19        - "10911:10911"
20        - "10912:10912"
21      environment:
22        - JAVA_OPT_EXT=-server -Xms128m -Xmx128m -Xmn64m
23      volumes:
24        - /home/rocketmq/broker/logs:/home/rocketmq/logs
25        - /home/rocketmq/broker/store:/home/rocketmq/store
26        - /home/rocketmq/broker/conf:/home/rocketmq/rocketmq-4.9.3/conf
27      command: sh mqbroker -c /home/rocketmq/rocketmq-4.9.3/conf/broker.conf
28      depends_on:
29        - rocketmq-namesrv
30      restart: on-failure
31    rocketmq-console:
32      image: apacherocketmq/rocketmq-dashboard:1.0.0
33      container_name: rocketmq-console
34      ports:
35        - "8280:8080"
36      environment:
37        - JAVA_OPTS=-Drocketmq.namesrv.addr=rocketmq-namesrv:9876 -
38          Dlogging.level.root=info -Dcom.rocketmq.sendMessageWithVIPChannel=false
39      depends_on:
40        - rocketmq-namesrv
41        - rocketmq-broker
42      restart: on-failure
```

4 启动与测试

前台启动服务

```
1 | docker-compose up
```

```
[root@localhost rocketmq]# ls  
broker  docker-compose.yml  rocketmq-4.9.3  rocketmq-all-4.9.3-bin-release.zip  
[root@localhost rocketmq]# docker-compose up
```

首次启动需要下载镜像文件，耐心等待即可，启动成功示意效果如下图

[illegible]


```
rocketmq-console | [2022-07-31 02:11:30.610] INFO Start time: 2449 ms
rocketmq-console | [2022-07-31 02:11:30.630] WARN Exception caught: mqAdminExt get broke
rocketmq-console | [2022-07-31 02:11:30.631] WARN Response [CODE: 1 DESC: The stats <TQ
```

连接控制台访问测试，访问地址<http://ip:8280/>，在集群查看是否可以看到Broker服务，如下图所示。

RocketMQ仪表板 运维 驾驶舱 集群 主题 消费者 生产者 消息 消息轨迹 更换语言 ▾										
集群: DefaultCluster ▾										
分片	编号	地址	版本	生产消息TPS	消费消息TPS	昨日生产总数	昨日消费总数	今天生产总数	今天消费总数	操作
broker-a	0(master)	192.168.220.128:10911	V4_9_3	0.00	0.00	0	0	0	0	状态 配置

到此可以看到一切正常，下一步可以切换到后台启动了Ctrl + c结束前台启动，使用下面命令进行后台启动

```
1 docker-compose up -d
```

```
rocketmq-console | For more information, please visit the url, http://rocketmq.apache.org/docs/faq/
^CGracefully stopping... (press Ctrl+C again to force)
[+] Running 3/3
 # Container rocketmq-console Stopped
 # Container rocketmq-broker Stopped
 # Container rocketmq-namesrv Stopped
canceled
[root@localhost rocketmq]# docker-compose up -d
[+] Running 3/3
 # Container rocketmq-namesrv Started
 # Container rocketmq-broker Started
 # Container rocketmq-console Started
[root@localhost rocketmq]#
```

使用docker ps指令查看后台启动情况

```
[root@localhost rocketmq]# docker ps | grep rocketmq
fe8b49d03947  apacherocketmq/rocketmq-dashboard:latest  "sh -c 'java $JAVA_O..." 21 minutes ago Up About a minute
tcp rocketmq-console
aeeco4e7e81e  apache/rocketmq:4.9.3 "sh mqbroker -c /hom..." 22 minutes ago Up About a minute
909/tcp, 9876/tcp, 0.0.0.0:10911-10912->10911-10912/tcp, :::10911-10912->10911-10912/tcp rocketmq-broker
26d70a9f18a7  apache/rocketmq:4.9.3 "sh mqnamesrv" 22 minutes ago Up About a minute
9876->9876/tcp, 10911-10912/tcp rocketmq-namesrv
```

因为容器已经创建，后续可以使用stop和start命令来控制容器的停止与启动。

```
[root@localhost rocketmq]# docker-compose stop
[+] Running 3/3
 # Container rocketmq-console Stopped
 # Container rocketmq-broker Stopped
 # Container rocketmq-namesrv Stopped
[root@localhost rocketmq]# docker-compose start
[+] Running 3/3
 # Container rocketmq-namesrv Started
 # Container rocketmq-broker Started
 # Container rocketmq-console Started
```

5 扩展阅读

docker安装常见问题解决参考

https://blog.csdn.net/apple_csdn/article/details/125321157

集群搭建参考

https://blog.csdn.net/apple_csdn/article/details/125296236

https://blog.csdn.net/apple_csdn/article/details/125284438